

Small-Scale SOC Simulation – Project Plan

1. Project Overview

This project is a Small-Scale Security Operations Center (SOC) Simulation designed to replicate the environment and responsibilities of a Junior Cybersecurity Analyst. It will involve deploying a SIEM platform (Security Onion or Wazuh) within a virtualized lab environment using UTM on a MacBook, connecting multiple endpoint virtual machines via agents, and simulating various security events for detection, analysis, and incident response practice.

2. Problem Statement

Organizations face continuous cyber threats and require skilled analysts to detect, analyze, and respond to incidents in real time. However, new analysts often lack practical experience working with SIEM platforms, endpoint monitoring, and incident response workflows. This project addresses that gap by creating a realistic, self-contained SOC environment where security events can be generated and analyzed, preparing for real-world junior analyst responsibilities.

3. Project Objectives

- Set up a fully functional SIEM (Security Onion or Wazuh) in a virtual lab.
- Deploy multiple endpoint VMs and configure endpoint agents.
- Collect and analyze logs from diverse sources (Windows, Linux, firewall).
- Simulate attacks and security incidents.
- Create custom detection rules for suspicious activity.
- Document incidents in a professional Incident Response Playbook.
- Demonstrate findings through dashboards, reports, and a final walkthrough.

4. Scope

In-Scope:

- SIEM deployment in UTM.
- Multiple endpoint systems (Windows + Linux).
- Open-source-only tools.
- Basic attack simulation and log ingestion.
- Documentation of environment, rules, and incidents.

Out-of-Scope:

- Integration with live enterprise networks.
- Handling real malware (only safe test files allowed).
- Cloud-based deployments.

5. Tools & Technologies

- **Virtualization:** UTM for macOS
- **SIEM:** Security Onion or Wazuh
- **Endpoint Agents:** Wazuh Agent, Beats (Winlogbeat/Filebeat), Syslog
- **Operating Systems:** Windows 10/11 Eval, Ubuntu Linux Server/Desktop
- **Attack Simulation:** Nmap, EICAR test file, basic brute-force scripts
- **Documentation:** draw.io / Lucidchart, Markdown, PDF reports

6. Timeline

Month 1 – Environment Setup

Goal: Functioning SIEM with connected endpoints and log flow.

- Install UTM on Mac and configure virtual networking.

- Deploy Security Onion or Wazuh in a VM.
- Create at least 2 endpoint VMs (Windows + Linux).
- Install and configure endpoint agents.
- Verify log ingestion.
- Document setup and architecture.

Month 2 – Detection & Simulation

Goal: Create alerts and simulate real-world attacks.

- Write 3–5 custom detection rules.
- Simulate brute force, port scans, and malware download attempts.
- Tune detection rules to reduce false positives.
- Start initial incident response documentation.

Month 3 – Incident Response & Final Deliverables

Goal: Produce a full incident response package and present findings.

- Create an Incident Response Playbook.
- Build SIEM dashboards for monitoring.
- Produce final incident reports with screenshots.
- Record a walkthrough/demo video.
- Submit final project document.

7. Month 1 – Week-by-Week Plan

Week 1:

- Install UTM on Mac.

- Download ISOs for SIEM and endpoint OS.
- Create base network diagram.

Week 2:

- Deploy SIEM VM (Security Onion/Wazuh).
- Configure basic settings (IP, networking, storage).

Week 3:

- Create endpoint VMs (Windows + Linux).
- Configure host-only or NAT networking in UTM.

Week 4:

- Install endpoint agents.
- Verify logs are visible in SIEM.
- Test simple log events (failed logins, command history).

8. Deliverables

- **Technical Documentation** – Environment setup guide with screenshots.
- **Network Diagram** – Showing SIEM, endpoints, and log flow.
- **Detection Rules** – List of custom rules and their purpose.
- **Incident Response Playbook** – Steps for responding to detected threats.
- **Final Dashboard** – SIEM dashboard with active monitoring views.
- **Final Presentation/Demo Video** – Walkthrough of lab setup, detection, and response.

9. Success Criteria

- All endpoint agents report to the SIEM.
- Custom rules trigger alerts for simulated attacks.
- Incident reports are clear, accurate, and professionally formatted.
- Final demo successfully shows the detection and response workflow.

10. Risks & Risk Mitigation

Risk	Impact	Likelihood	Mitigation Strategy
Performance limitations on MacBook when running multiple VMs	Slow SIEM response and delayed log ingestion	Medium	Allocate sufficient RAM/CPU to critical VMs, reduce endpoint count if necessary, stagger testing workloads
Networking configuration issues in UTM	Endpoints unable to communicate with SIEM	Medium	Use host-only/NAT networks with documented IP assignments; test connectivity early
SIEM or agent misconfiguration	No logs or incomplete log data	High	Follow setup documentation step-by-step; verify ingestion with sample logs before simulations
Detection rules too broad (false positives)	Analyst overwhelmed with alerts	Medium	Tune rules iteratively and test with controlled simulations

Limited time for
simulation/incident
response documentation

Incomplete
deliverables

Low

Follow week-by-week plan
and document as you go

11. Future Considerations

- **Scaling the Lab** – Expand the environment to include additional endpoints, network segments, or specialized log sources (e.g., IDS, firewalls, web servers).
 - **Advanced Attack Simulation** – Incorporate tools like Metasploit, Caldera, or Atomic Red Team for realistic adversary emulation.
 - **Cloud Integration** – Experiment with hybrid SOC setups by integrating cloud-based log sources (AWS, Azure, GCP) into the SIEM.
 - **Automation & SOAR** – Add a Security Orchestration, Automation, and Response layer for automated incident handling.
 - **Team Collaboration** – Enable multiple analysts to interact with the lab simultaneously to simulate SOC teamwork.
 - **Certification Prep** – Use the environment to practice scenarios for certifications like CompTIA Security+, CySA+, or Splunk Core Certified User.
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